

DATA QUALITY & SURVEY RELIABILITY AUDIT

Answers QA1. What share of ER visits has a recorded patient satisfaction score, and is the missing data random or systematically biased? and QA2. Does the dataset contain structural data-quality issues — duplicate records, impossible ages/wait times, or inconsistent labels?

Total visits

9216 Row count of the cleaned dataset.

Duplicate Patient IDs found

0 Sum of the Duplicate Patient Id Flag column; 0 = no duplicate patient records.

Age outliers (age < 0 or > 110)

0 Sum of the Age Outlier Flag column.

Wait-time outliers (< 0 or > 300 min)

0 Sum of the Waittime Outlier Flag column.

Visits with a satisfaction score (responders)

2517 Sum of the Has Satisfaction Score flag.

Visits missing a satisfaction score

6699 Total visits minus responders.

Survey response rate

27.3% Responders / total visits — formatted as %.

Non-response bias check: responders vs. non-responders

Survey Group	Visits	Avg Wait Time	Admission Rate
Responded	2517	35.4	49.1%
No response	6699	35.2	50.4%

Conclusion

Only ~27% of visits carry a satisfaction score; wait time and admission rate are close between responders and non-responders (see Python notebook for the formal significance test), so the low response rate itself — not a detectable bias — is the data-quality issue to flag when reporting the satisfaction KPI.

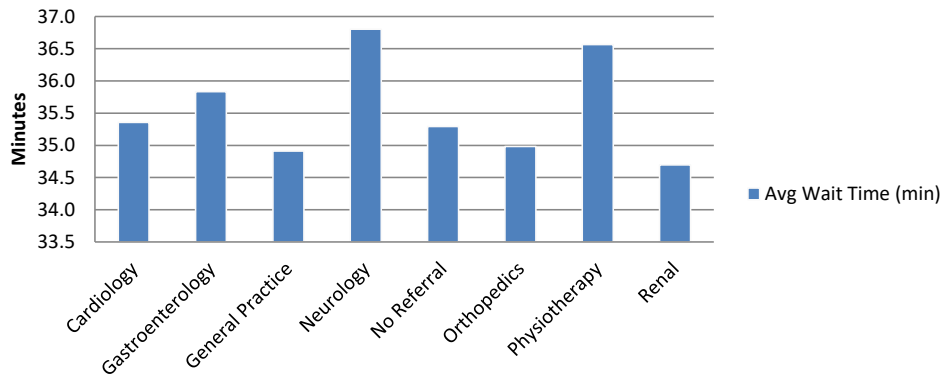
DEPARTMENT REFERRAL: WAIT TIME & SATISFACTION

Answers QB3 Which department referrals combine the longest wait times with the lowest satisfaction — where should resources be reallocated?

— ranks departments by average wait time and average satisfaction to flag where to reallocate resources.

Department	Visits	Avg Wait Time (min)	Avg Satisfaction (responders)	Admission Rate
Cardiology	248	35.4	5.14	49.2%
Gastroenterology	178	35.8	5.80	50.0%
General Practice	1840	34.9	5.06	48.2%
Neurology	193	36.8	5.28	50.3%
No Referral	5400	35.3	4.95	50.7%
Orthopedics	995	35.0	4.86	50.1%
Physiotherapy	276	36.6	4.99	49.6%
Renal	86	34.7	4.57	53.5%

Average Wait Time by Department



DAY x HOUR STAFFING GAP HEATMAP

Answers QD2 Where does demand most exceed service capacity, and where should the next shift's staff be placed?

— % of visits missing the 30-minute target, by day of week and 2-hour arrival window. Darker = bigger staffing gap.

Hour	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
00-02	56.1%	57.9%	58.8%	56.6%	54.9%	60.2%	55.1%
02-04	68.5%	57.9%	62.5%	61.0%	61.9%	60.9%	61.2%
04-06	62.4%	68.0%	55.8%	56.7%	58.2%	61.4%	60.9%
06-08	65.5%	52.0%	51.4%	56.5%	60.6%	62.7%	59.7%
08-10	63.8%	61.6%	64.5%	56.6%	55.9%	61.3%	57.0%
10-12	63.6%	65.1%	62.2%	64.9%	52.5%	58.7%	61.6%
12-14	54.8%	66.4%	59.0%	55.2%	54.2%	61.8%	55.9%
14-16	54.5%	50.0%	53.0%	60.4%	60.8%	59.1%	60.5%
16-18	63.5%	56.0%	59.8%	52.1%	62.2%	58.6%	55.7%
18-20	55.3%	60.7%	59.6%	57.1%	53.2%	54.0%	60.0%
20-22	62.7%	57.1%	57.7%	46.6%	57.3%	60.0%	64.5%
22-24	63.2%	52.5%	67.3%	70.7%	62.6%	69.3%	64.3%

Reference: 'Target Missed' = Patient Waittime > 30 minutes (Waittime Status column).

Insight: the darkest cells mark the day/hour windows where the current staffing level most consistently fails the 30-minute target — the direct i